

A BLOCKCHAIN SECURITIES TOKENIZATION FRAMEWORK
AND
A STOCHASTIC ANALYSIS OF INTERBANK PAYMENT NETWORKS

by

Reina Ke Xin Li

A thesis submitted in conformity with the requirements
for the degree of Master of Applied Science
Department of Electrical and Computer Engineering
University of Toronto

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2025

Abstract

This thesis explores two complementary areas in finance: securities tokenization and interbank payments. First, we propose a tokenization framework extending decentralized finance (DeFi) benefits—accessibility, transparency, efficiency—to real-world securities. While DeFi mechanisms like liquidity pools complicate securities entitlements (e.g., dividends, voting), our solution overcomes this by combining fungible tokens with off-chain accounting and separate smart contracts for entitlements. Implemented on Ethereum, it saves 27% in costs compared to alternatives and supports various securities including stocks and bonds. We also confirm compatibility with liquidity logic in 90% of Ethereum pools. Second, we model Real Time Gross Settlement (RTGS)-based Large Value Payment Systems (LVPS) using queuing theory. RTGS is the most common mechanism for LVPS due to its instant settlement and low risk, but it has high liquidity requirements. Our model yields closed-form solutions for network performance and offers insights into liquidity requirements to aid the design of liquidity saving mechanisms.

Acknowledgements

I would like to extend my gratitude to my supervisor, Professor Andreas Veneris, for his insightful guidance throughout the course of my Master's program. His thoughtful feedback and encouragement were instrumental in shaping this work. His support presented me with invaluable opportunities that led to exciting research and meaningful collaborations. I am also grateful to my other collaborators, Srisht Fateh Singh, Professor Andreas Park, Professor Darrell Duffie, and Professor Peter Glynn for their ideas, support, and contributions, without which this work would not have been possible. I want to further thank Professor Andreas Park and Professor Fan Long for serving on my thesis defense committee and providing their time and valuable feedback, and Professor Ali Hooshyar for chairing my MASc Thesis Oral Examination. I would like to acknowledge the financial support provided by the University of Toronto Department of Electrical and Computer Engineering and the Ontario Graduate Scholarship. Finally, I sincerely thank my friends and family for their support.

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Chapter 1

Introduction

The global need for secure, efficient, and scalable financial infrastructure is becoming increasingly important due to the high volumes and values processed by financial systems today—McKinsey & Company reports 3.4 trillion payment transactions worth \$1.8 quadrillion USD processed globally in 2023 [19], while the Bank for International Settlements (BIS) estimates a global daily average of \$7.5 trillion USD of foreign exchange transactions in 2022, 30 times the global daily gross domestic product (GDP) [18]. Furthermore, there is a strong push for self-sovereignty, self-custody, and accessibility in finance to address the inequities inherent to traditional arrangements. The response is a global effort to innovate in the financial technology space, which underpins the evolution of financial systems worldwide.

From decentralized blockchain technology to highly centralized large-value interbank payment systems, the infrastructure underlying the global economy is increasingly taking advantage of programmability and automation. These technologies aim to improve efficiency, reduce costs, provide accessibility, and mitigate risk. The BIS expects “the digitalisation of countries’ payment ecosystems” driven by fast payments [40], and emphasizes that “payment systems [are being redesigned] to make them faster, cheaper and easier to use [and that] consumers have come to expect continuous improvements in the convenience and safety of payment services” [17]. However, innovations in financial technologies also introduce complex design and integration challenges, particularly when adapting traditional financial mechanisms to new technological environments.

The past decade has seen the emergence and rise of decentralized systems, which rely on distributed ledger technologies (blockchain) and smart contracts to enable peer-to-peer transactions without intermediaries. The developments in the decentralized space make these systems progres-

sively more relevant to financial use cases—beyond the simple transfers of value they were originally designed for. For instance, in *decentralized finance* (DeFi) ecosystems, decentralized exchanges were created to facilitate the trading of assets within the ecosystem (*i.e.*, on-chain) [1]. These types of DeFi constructs lend themselves to the pursuit of real-world asset *tokenization*. Tokenizing traditional financial instruments such as stocks and bonds offers the potential to democratize access, eliminate intermediaries, and bring transparency to capital markets. It could also increase liquidity, reduce settlement times, and lower operational costs associated with traditional securities operations [51].

Yet, realizing these benefits in practice remains difficult. Namely, existing tokenization schemes fail to capture complex securities contract terms that offer their owners more than just ownership rights. The dynamic features of some financial instruments—such as dividends, voting rights, and interest—complicate their tokenization, particularly in their integration with the DeFi ecosystem. For instance, the decentralized nature of DeFi constructs can make the ownership of assets ambiguous or difficult to determine when it comes to exercising contractual rights [52]. Furthermore, assets may have rights that accrue over time and vary based on holding periods (*e.g.*, bond interest), a case that needs to be handled accordingly [14]. [72] acknowledges this shortcoming in the discussion of their tokenization solution for dividend-paying stocks. Additionally, financial institutions tend to be hesitant when it comes to the openness of DeFi—their solutions ([72, 3, 65, 24, 67, 15, 48, 57]) often curtail DeFi operability in favour of strict regulability. On the scalability side, blockchain operations can be expensive, posing another limitation to many solutions (*e.g.*, [35]).

Meanwhile, centralized financial infrastructures face their own challenges. For instance, research and development on *Large Value Payment Systems* (LVPS) has been long ongoing. These systems settle high-value transactions (such as those supporting securities settlement and foreign exchange) between financial institutions and are essential for economic function and monetary policy [17, 39]. Globally, the dominant settlement mechanism for large value payments is *Real-Time Gross Settlement* (RTGS), which offers strong guarantees around payment finality and credit risk [8]. However, it requires significant liquidity as participants need to maintain sufficient settlement reserves in real-time.

Another common mechanism is *Deferred Net Settlement* (DNS), where payments are netted and settled at the end of a clearing cycle, significantly reducing the liquidity burden on its participants. However, DNS is vulnerable to credit risk due to delayed settlement, the costs of which must be born by participants or, in some cases, the system operator [54].

As a result, central banks and other payment system operators have responded to these concerns by implementing *liquidity saving mechanisms*—algorithms that match and net offsetting payments

throughout the day, eliminating the need for liquidity to settle both directions of netted payments while still ensuring fast settlement times. Much work has been done to empirically measure the performance improvements of these mechanisms [27, 53, 26]. To complement this data-driven analysis, there is a need to analytically model and understand the stochastic dynamics of payment systems and their performance.

1.1 Contributions

This thesis addresses the aforementioned challenges in both the decentralized and the centralized financial systems by presenting a practical mechanism for the former and a theoretical model for the latter. The two main contributions of this thesis are:

- **Blockchain Meets Securities: A Scalable Tokenization Framework.** We propose a comprehensive framework for tokenizing traditional securities (e.g., stocks, bonds) on Ethereum that integrates seamlessly with DeFi protocols. This framework addresses critical challenges such as ownership attribution and rights distribution for tokenized assets in DeFi ecosystems. The proposed solution comprises:
 - A Stock Token Contract that creates liquid, fungible representations of securities compatible with general DeFi protocols.
 - An Off-Chain Accounting procedure that computes token ownership and rights allocation even for assets pooled in smart contracts, without incurring high on-chain costs. This procedure uses publically available blockchain transaction data to account for tokens and interactions with DeFi protocols in a transparent manner.
 - A Rights Redemption Contract that enables on-chain redemption of dividends, voting rights, or other entitlements based on verified off-chain ownership accounting. This contract relies on asymmetric key signatures to enable the secure, verified, and low-cost release of ownership entitlements.
 - An extension to handle coupon-paying bonds that uses a holding time-based ownership accounting model to allocate accrued bond interest between transfers. This extension is a modification of the Off-Chain Accounting procedure to analyze token ownership on a block-by-block basis. The approach also offers better risk sharing for coupon defaults.

The methodology generalizes to any asset with time-dependent rights that can be represented on-chain (*e.g.*, voting rights, mergers, dividends, coupons, *etc.*) and results in a flexible to-

kenization system that maintains DeFi interoperability. It results in a 27% reduction in gas cost compared to other tokenization solutions for dividend-paying assets and is compatible with over 90% of DeFi liquidity pool protocols.

- **Modelling RTGS-based Interbank Payment Networks.** We develop a stochastic model for RTGS systems to analyze liquidity requirements and payment throughput. The proposed model maps RTGS payment flows to a Gordon-Newell queuing network and uses existing queuing theory results to analytically derive closed-form equilibrium characteristics and extends them to handle randomly sized payments. The model shows that, for RTGS systems:

- Networks with balanced payment flows result in simple closed-form solutions. Such balanced networks behave like fully homogeneous networks, even though they may have heterogeneous routing characteristics and payment dynamics.
- Long-run outcomes depend only on the structure and flow dynamics of the network rather than on its initial conditions. The ratios of key parameters—the routing characteristics, payment arrival rates, and transaction sizes—emerge as critical drivers of network’s steady-state behaviour.

This model-based approach offers a complementary perspective to empirical analysis, enabling the examination of system dynamics under controlled assumptions. It provides a theoretical basis that can help inform effective payment system design, with respect to liquidity efficiency and throughput.

1.2 Outline

The remaining chapters of this thesis are organized as follows: Chapter 2 provides a high-level background of the relevant concepts in blockchain, tokenization, payments, and modelling, Chapter 3 presents the first contribution, a tokenization framework, Chapter 4 presents the second contribution, an interbank payments network model, and Chapter 5 summarizes the contributions of this thesis and outlines directions for future work.

Chapter 2

Background

This section provides a background on securities tokenization and the relevant technologies, as well as interbank payment systems and stochastic modelling.

2.1 History and Basics of Blockchain

Blockchain is a decentralized, distributed ledger technology that enables secure, transparent, and immutable transactions. It consists of a chain of blocks, each containing a batch of timestamped transactions linked using cryptographic hashes. This structure forms an immutable ledger, as illustrated in Figure 2.1. Unlike centralized systems, blockchains typically operate on peer-to-peer networks, eliminating the need for trusted central authorities. Blockchains are most famously associated with cryptocurrencies—digital tokens recorded and transacted on the ledger.

2.1.1 Before the Launch of Bitcoin

The invention of the modern blockchain is built upon a large body of work among cryptographers and computer scientists in the late 20th century. The first-known blockchain-like protocol is e-cash

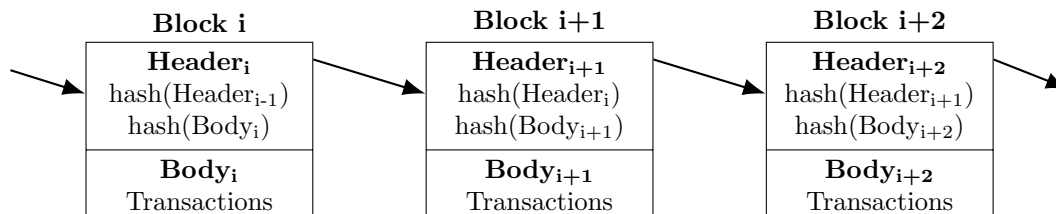


Figure 2.1: Simplification of a blockchain, where blocks are chained by containing the hash of the previous block's header.

by David Chaum in 1982, who proposed a design for decentralized computing among mutually suspicious computer groups using cryptographic techniques [21]. In 1991, Haber and Stornetta published a solution for time-stamping digital documents in a manner that prevents back-dating, forward-dating, and tampering using linking/chaining [36]. Back, in 2002, published a whitepaper introducing the concept of proof-of-work, originally used to combat denial-of-service attacks and spam emails [5]. Proof-of-work was later adapted to be a decentralized consensus mechanism for blockchain.

2.1.2 Bitcoin Blockchain: Proof-of-Work Consensus

Blockchain technology itself is most famously credited to the anonymous Satoshi Nakamoto in the 2008 Bitcoin whitepaper [59], which cites Haber and Stornetta (1991) and Back (2002). The actual Bitcoin network, recognized as the first blockchain protocol, was then launched in 2009, enabling the decentralized transacting of the digital Bitcoin cryptocurrency.

The technology relies on cryptographic hashing and consensus mechanisms to validate and add blocks of transactions in a decentralized manner. The Bitcoin blockchain’s consensus mechanism is a proof-of-work mechanism, similar to that proposed by Back in [5]. Network participants—called miners—solve computationally-intensive problems, a process mining, to gain the right to add new blocks to the chain. A miner builds a block of valid transactions, solves the proof-of-work puzzle, and proposes (broadcasts) their block (embedded with the solution) to the network. Other miners verify the correctness of the proposed block and “vote” to accept the block by building and mining the next block on top of it. In the case of network forks, which may occur when miners receive blocks at different latencies or vote for different blocks, the chain that becomes longest is deemed correct.

The computational expense of mining makes it economically infeasible for attackers. For instance, an individual attempting to rewrite the blockchain (*e.g.*, to spend a token twice—called double spending) needs to outpace the combined mining power of the rest of the network in order to extend a chain containing their fraudulent block past the honest chain.

2.1.3 Challenges and Continued Development of Blockchain

The proof-of-work process ensures security but comes at a high energy cost. In 2022, Bitcoin mining consumed over 200 TWh globally—approximately 13% of the UK’s total electricity con-

sumption [61].¹ To address these concerns, alternative consensus mechanisms such as *proof-of-stake* (PoS) have been developed.

Blockchain technology faces other issues such as scalability and costs, centralization, regulatory uncertainty, and trade-offs between privacy, anonymity, auditability, and accountability. Ongoing research continues to address these while developing new use cases for blockchain in order to revolutionize traditional industries. Since its inception, the use of blockchain has evolved beyond cryptocurrencies: it has applications in various industries, including finance, healthcare, voting and governance systems, and digital identity verification. Smart contracts, pioneered by Ethereum, further expanded blockchain’s capabilities by enabling programmability.

2.2 Ethereum Blockchain

Ethereum is a blockchain with capabilities beyond token transactions. Proposed in 2013 by Buterin, it enables Turing-complete programmability via *smart contracts*—code written in Solidity language that allows arbitrary states and state transitions to be stored and executed on the blockchain [20]. Ethereum users pay for the execution of smart contracts (known as gas fees) using the blockchain’s native cryptocurrency, Ether (ETH).² ETH is also used as an investment asset and the native medium of exchange.

Turing-complete, open-access programmability is what positions Ethereum as a technology with the potential to disrupt traditional systems. Smart contracts give the ability to anyone to build decentralized applications on Ethereum, a revolutionary feature given the restrictions traditionally in place around the development and deployment of financial technologies. Examples of decentralized applications include token exchanges, financial instruments like escrows, decentralized governance, and self-sovereign identity.

While Ethereum initially used proof-of-work consensus, it transitioned to proof-of-stake in 2022 to improve its energy efficiency. Proof-of-stake relies on participants (called validators) staking ETH on the correctness of their contributions to the blockchain in order to secure the network, rather than competing through mining.³ Further scalability and efficiency solutions have been introduced and are being developed, including layer 2 solutions—technologies like optimistic and zero-knowledge rollups that help reduce gas fees and increase transaction throughput.⁴

Ethereum has seen use cases in many industries, such as finance (*e.g.*, decentralized finance),

¹Bitcoin energy consumption estimates: <https://digiconomist.net/bitcoin-energy-consumption>

²See <https://ethereum.org/en/developers/docs/gas/>.

³See <https://ethereum.org/en/developers/docs/consensus-mechanisms/pos/>.

⁴See <https://ethereum.org/en/developers/docs/scaling/>.

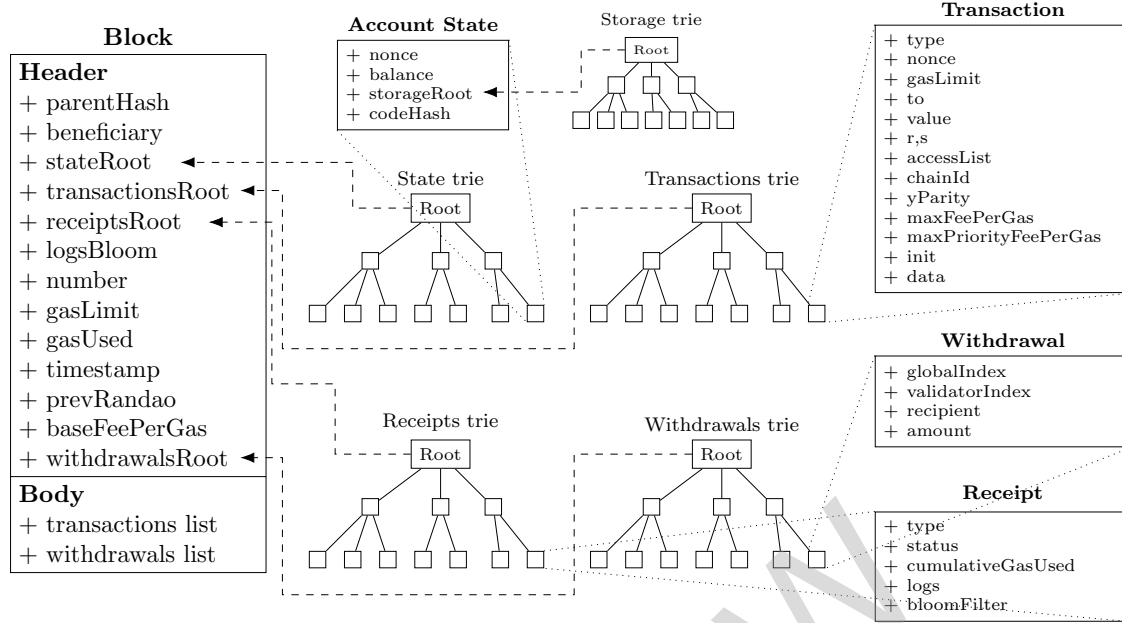


Figure 2.2: Ethereum block structure.

digital art (*e.g.*, non-fungible tokens (NFTs)), gaming and Web 3.0 (*e.g.*, metaverse), and enterprise blockchain solutions (*e.g.*, supply chain tracking, identity management).

2.2.1 Ethereum Nodes and Data

An Ethereum validator can run a *full node*, which stores every block. A block's data includes the hash of the preceding block (chaining), the hash of the state trie root, information regarding the consensus mechanism, the transactions, and the hash of the transaction receipts (*e.g.*, gas used by and events emitted by the execution of smart contracts) trie root [70]. Specifically, the Ethereum block structure is depicted in Figure 2.2. A block can be thought of as the instructions for the blockchain's state transition—instead of storing each state, full nodes only store these instructions. This data allows a full node to regenerate the state at any block in the history of the blockchain by locally executing the transactions sequentially. Full nodes also store the states of the last 128 blocks in order to easily access these states in case of block reorganizations (*e.g.*, network forks).

Meanwhile, a network participant can choose to run an *archive node*, which stores the state at every block in the blockchain's history. Archive nodes allow for faster retrieval of historical state information at the cost of heavy storage space requirements. While a full node typically only requires up to 1 TB of storage, an archive node can require up to 12 TB.⁵

Finally, Ethereum participants can also choose to run a *light node*, which only downloads and

⁵See <https://ethereum.org/en/developers/docs/nodes-and-clients/>.

verifies block headers. Light nodes may request and verify data from full or archive nodes against the chain of headers, meaning they can access blockchain data reliably with much less overhead.⁶

2.2.2 Asymmetric Key Cryptography on Ethereum

Asymmetric key algorithms are a type of encryption system that uses a pair of related keys: a public key and a private key (hence it is also called public-key cryptography). The public key is shared openly and is used for encryption and verification, while the private key is kept secret and is used for decryption and signing. This asymmetry allows secure communication between parties without having to share a common secret key beforehand.

The *Elliptic Curve Digital Signature Algorithm* (ECDSA) is one such algorithm that is used to create and verify digital signatures. Its basis is elliptic curve cryptography, which provides security guarantees while using short keys [60]. This makes ECDSA efficient in terms of computation and storage, which is suitable for resource-constrained environments, such as blockchain networks.

Ethereum relies on ECDSA to ensure account and transaction security.⁷ Accounts on Ethereum own an ECDSA public and private key pair, the former of which is hashed to generate an address—the account’s public identifier. When a user initiates a transaction, they sign it with their private key to prove ownership of the account making the transaction (without revealing their private key). Their corresponding public key can be used to verify the authenticity of the signature, ensuring that the transaction was indeed signed by the rightful account owner and that the transaction details were not tampered with.

2.2.3 Ethereum Tokens

Tokens on Ethereum are implemented by smart contracts, enabling programmability, integrability, and security. Most often, tokens are implemented following *Ethereum Request for Comment* (ERC) standards, which enforce a common standardized interface for all tokens of a certain type. This makes it easy for developers to design applications that are operable with generic tokens and for users to interact with different tokens using the same tools. Token smart contracts keep track of token owners, allow secure transferring and burning, grant owners the ability to share control with others, and authorize permissioned minting.

Two of the most common Ethereum token types are defined by the ERC20 and ERC721 stan-

⁶See <https://ethereum.org/en/developers/docs/nodes-and-clients/light-clients/>.

⁷See <https://ethereum.org/en/glossary/#ecdsa>.

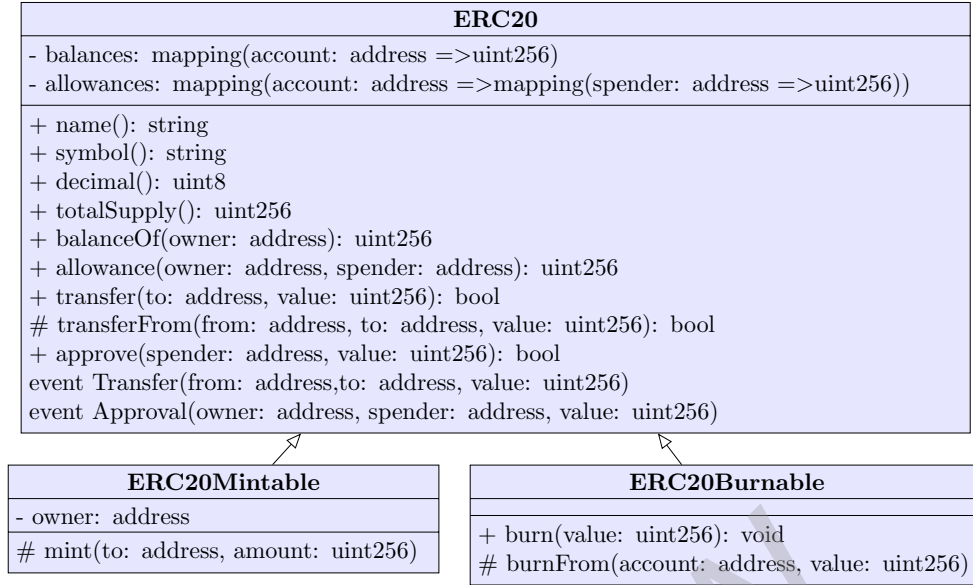


Figure 2.3: ERC20 smart contract class and extensions by OpenZeppelin.

dards: fungible and non-fungible tokens, respectively. The former, depicted in Figure 2.3,⁸ requires a smart contract to record the balances of each token owner. These tokens behave similarly to cash, in that individual tokens are interchangeable. Meanwhile, smart contracts for ERC721, depicted in Figure 2.4,⁹ assign unique identifiers (a number) to each token and map token identifiers to their owners. ERC721 tokens are often used to digitally represent unique assets, like identities, certificates, badges, or artwork.

There are a myriad of other ERC token standards, including semi-fungible tokens (ERC3525), multi-tokens (ERC1155), consumable tokens (ERC2135), tokenized vaults (ERC4626), internet-of-things device tokens (ERC4519), and compliant transfer security tokens (ERC3643).¹⁰

2.3 Decentralized Finance

Decentralized Finance (DeFi) is the financial ecosystem built on blockchains (most notably, on Ethereum) with the objective of creating a transparent and accessible financial system that anyone can use and on which anyone can develop. Unlike traditional financial systems, where intermediaries are required in order to perform transactions, DeFi protocols allow users to interact directly with decentralized applications (dApps) to access financial services (*e.g.*, lending, borrowing, trading, and

⁸Based on the OpenZeppelin implementation, at <https://github.com/OpenZeppelin/openzeppelin-contracts/tree/master/contracts/token/ERC20>.

⁹Based on OpenZeppelin implementation, at <https://github.com/OpenZeppelin/openzeppelin-contracts/tree/master/contracts/token/ERC721>.

¹⁰For a list of ERC standards, see <https://eips.ethereum.org/erc>.

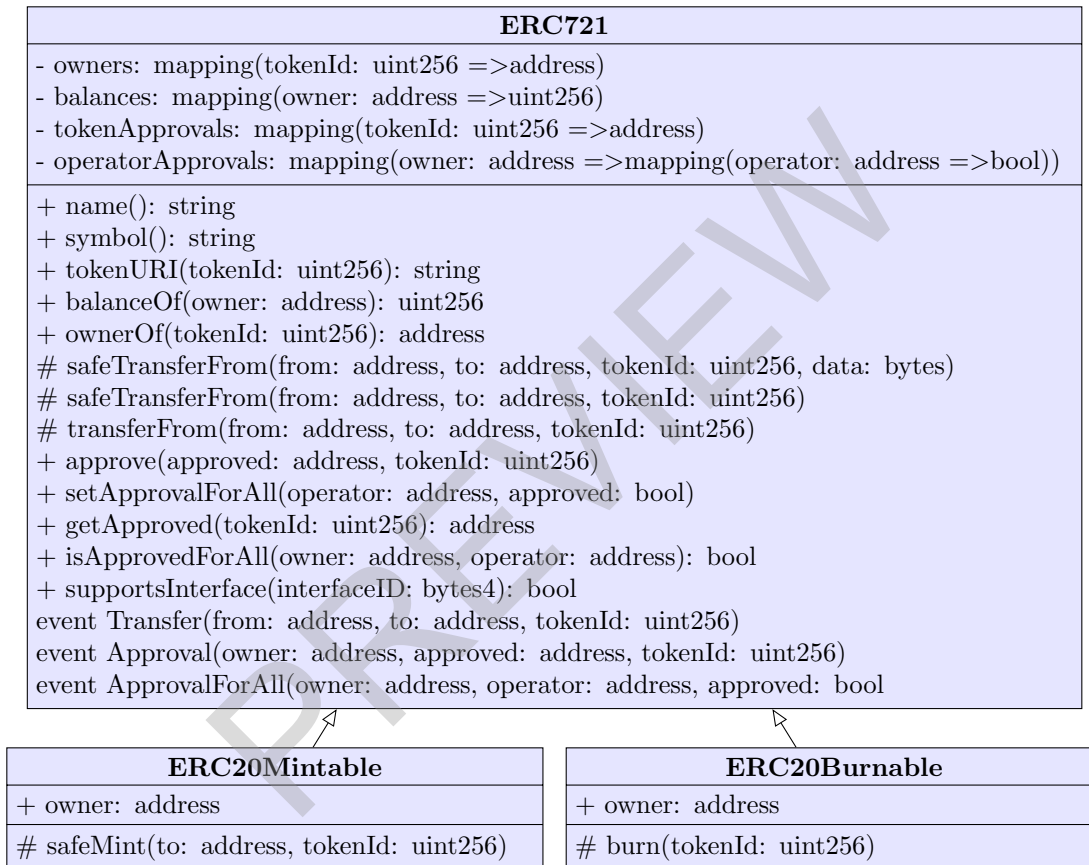


Figure 2.4: ERC721 smart contract class and extensions by OpenZeppelin.